

RFID Side Channel Analysis for IoT Devices

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Abstract— Side channel attacks on RFID devices present a significant threat by enabling attackers to extract secrets through indirect information such as power consumption or electromagnetic (EM) emissions. Existing approaches to RFID security often focus on cryptographic protocol analysis, but may overlook vulnerabilities exposed by physical layer leakage. This paper applies side channel analysis techniques to MIFARE Classic cards using a Proxmark reader, RTB2002 oscilloscope, and probes for both power and EM emissions. Controlled authentication sequences with known keys and data were repeated and measured, generating CSV samples for power and EM signals across multiple RFID memory blocks. These datasets will be analysed using leakage modelling and side channel evaluation metrics to assess the possibility of key retrieval. The findings aim to demonstrate practical risks in legacy RFID deployments and inform the design of more secure contactless systems.

I. INTRODUCTION (HALF PAGE)

MIFARE Classic cards represent one of the most widely used contactless smart card technologies, used in transport ticketing, building access control, and payment systems worldwide. These cards rely on the proprietary Crypto1 stream cipher to secure mutual authentication and data transmission. Although Crypto1 has been comprehensively broken through various cryptanalytic techniques, including brute-force and algebraic attacks, the potential for additional key-related information leakage via physical side channels remains underexplored. Side channel attacks exploit unintended information leakage through physical factors such as power consumption, electromagnetic (EM) emissions, timing variations, or acoustic emissions. Unlike traditional cryptanalytic attacks that target mathematical weaknesses in algorithms, side channel attacks focus on implementation that is dependent on physical characteristics that may reveal secret keys through statistical analysis of measurement data.

Previous research has demonstrated successful side channel attacks against various cryptographic implementations, particularly through Differential Power Analysis (DPA) and Correlation Power Analysis (CPA) techniques. These methods exploit correlations between measured power consumption and intermediate cryptographic values, often using leakage models such as Hamming weight or Hamming distance of processed data.

This paper investigates whether MIFARE Classic authentication commands produce measurably different power and electromagnetic emissions depending on the Crypto1 key being used. Power consumption and electromagnetic emission measurements, combined with statistical analysis and correlation based leakage model evaluation, to determine the feasibility of key discrimination through side channel analysis. The results demonstrate significant electromagnetic leakage correlating with key Hamming weight, providing evidence of exploitable key dependent side-channel signatures in MIFARE Classic implementations.

II. PRIOR WORK (HALF PAGE)

To Do: prior work

III. TECHNICAL DESCRIPTION (TWO PAGES)

This paper investigates side channel leakage from MIFARE Classic RFID cards by measuring power consumption and EM emissions during authentication. The aim is to test whether secret key or block data processed by the Crypto1 stream cipher produce measurable physical signatures exploitable in practical attacks.

A. Experimental Hardware Setup

The hardware used in this experiment consisted of a Proxmark3 device, a Rohde & Schwarz RTB2002 digital oscilloscope, and a MIFARE Classic 1K RFID card.

The Proxmark3 was used to communicate directly with the RFID card and issue commands such as key authentication and block writing. It also acted as the trigger source for synchronising oscilloscope captures.

A probe was connected to the Proxmark to capture the power used when executing the commands.

A high sensitivity, near field EM probe (NAE Hprobe 15) was used to capture side channel emissions during card communication.

The RTB2002 oscilloscope was used to record traces from these two analogue channels, measuring EM and power related signals during MIFARE operations.

The MIFARE Classic 1K card served as the target of the analysis. It was placed directly on the Proxmark, which was attached to a sensing probe and the EM probe placed directly on top of the MIFARE card during trace collection to ensure consistency. Each test sequence involved sending the same command multiple times to the same block or sector, with the hardware setup remaining in the same physical position throughout.

Image Place Holder

Fig 1: Hardware Setup

B. Key Configuration

Three distinct Crypto1 keys were programmed into different sectors of the MIFARE Classic card to enable comparative analysis:

Block 1 Key: 434A9734E087

Block 2 Key: 010203040506

Block 3 Key: A0A1A2A3A4A5

Keys were selected at random. The Proxmark3 ‘hf mf wrbl’ command was used to program sector trailer blocks with the respective keys, followed by verification using ‘hf mf chk’ commands.

C. Data Collection

To gather consistent side channel data, a list of repeatable measurement steps were created. For each test, a single MIFARE Classic 1K card was

placed directly on the Proxmark with the EM probe placed on top of the card, which was connected to the oscilloscope. The card was not moved between repeated tests to minimise any variations in the data. The same authentication was sent from the Proxmark was repeated 15 times per block and key combination. Each command triggered a waveform capture on the oscilloscope, recording power, EM and timing data. The oscilloscope was configured to begin capturing as soon as the command was issued, using digital triggering based on signal activity to align each trace.

Waveforms were saved in CSV format with filenames clearly indicating the card, block, and repetition number (e.g., CARD1_01.CSV, CARD2_10.CSV). This ensured proper trace tracking and data labelling during analysis. All captures were carried out using the same sampling rate, voltage scale, and time base settings to ensure consistency across the entire dataset.

This list of steps allowed for fair comparisons between different keys and blocks, making it possible to evaluate whether observable variations in power or EM patterns could be linked to the specific key used.

IV. RESULTS OBTAINED (ONE PAGE)

A. Trace Characteristics

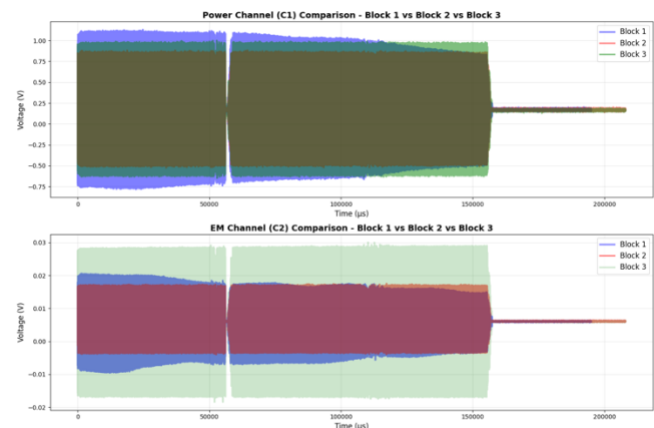


Fig 2: Overlay of averaged power (C1) and EM (C2) traces for all three keys during MIFARE Classic authentication

Figure 1 shows the averaged side channel traces for all three keys and blocks. Both power and EM measurements show identical timing structure, confirming that all traces successfully captured the

same ‘hf mf chk’ authentication protocol. From these traces, it was determined that the timings were almost identical but there was a visible difference in amplitude.

B. Statistical Analysis

Table 1 summarises the statistical comparisons of side channel features between each pair of keys. The mean and peak amplitudes in power (C1) and EM (C2) channels over the full 0–110 000 μ s window were analysed.

Pair	Channel	Metric	T-statistic	P-value	Cohen's d
Block1 vs Block2	C1	Mean	1.785	0.083	0.5754
Block1 vs Block2	C2	Mean	-1.927	0.0625	-0.6174
Block1 vs Block2	C1	Peak	5.21	0	1.8056
Block1 vs Block2	C2	Peak	5.737	0	1.8534
Block1 vs Block3	C1	Mean	5.611	0	1.7743
Block1 vs Block3	C2	Mean	0.225	0.823	0.0712
Block1 vs Block3	C1	Peak	14.979	0	4.7368
Block1 vs Block3	C2	Peak	-4.699	0.0001	-1.486
Block2 vs Block3	C1	Mean	4.222	0.0002	1.4051
Block2 vs Block3	C2	Mean	2.401	0.0218	0.7802
Block2 vs Block3	C1	Peak	2.511	0.0212	0.8857
Block2 vs Block3	C2	Peak	-6.418	0	-1.9577

Table 1: Statistical Test Results for Power (C1) and EM (C2) Channels

Peak amplitudes differ highly significantly for every key pair in both channel, with Cohen's d ranging from 0.886 to 4.737. Mean amplitudes also show significant differences in most comparisons, particularly for the power channel. These results confirm that both power and electromagnetic side channels leak key dependent information, and that peak based features show the greatest difference.

C. Hamming Weight Correlation Analysis

Block	HW
Block 1	21
Block 2	9
Block 3	19

Table 2: Hamming Weight of Keys per Block

Given the limitations of traditional statistical tests, correlation power analysis techniques were used to investigate whether side channel amplitudes correlate with the Hamming weight of the processed keys.

For each individual trace, 45 total, 15 per key, the peak voltage was extracted within the primary processing window, 0-150,000 μ s, for both channels. These peak values were then correlated

with their corresponding key Hamming weights using Pearson's correlation coefficient.

D. Leakage Model Validation

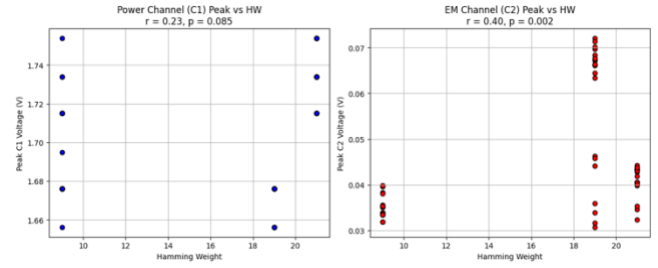


Fig 2: Correlation Between Peak and Hamming Weight For C1 and C2

The significant correlation observed in EM measurements ($p = 0.001$) provides strong evidence supporting the Hamming weight leakage model. Keys with higher Hamming weights consistently produce higher peak EM emissions during authentication processing.

The correlation strength ($r = 0.41$) indicates that approximately 17% of the variance in EM peak amplitudes can be explained by key Hamming weight alone. These findings show that there is a potentially large information leakage that could be used for key discrimination or recovery attacks.

V. ANALYSIS (ONE PAGE)

VI. CONCLUSION

REFERENCES

APPENDIX

Appendices, if needed, appear before the acknowledgement.